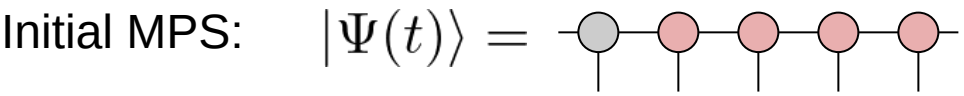


Class 11

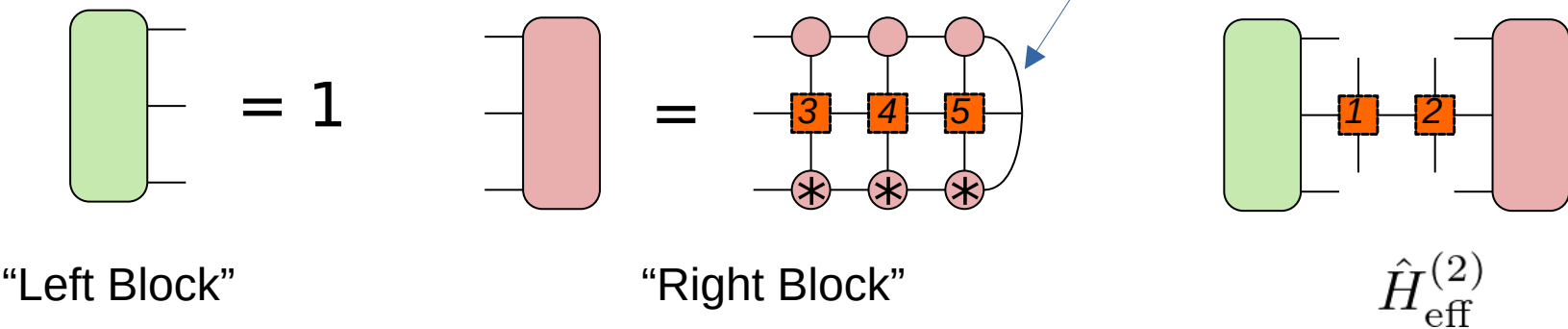
TDVP demo & Lindbladian Systems

Miklós Antal Werner
HUN-REN Wigner RCP

2-site TDVP recap (MatLab demo on GitHub)

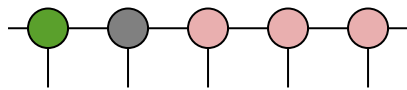
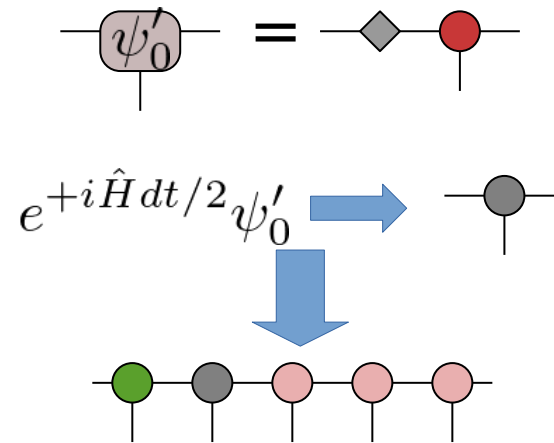
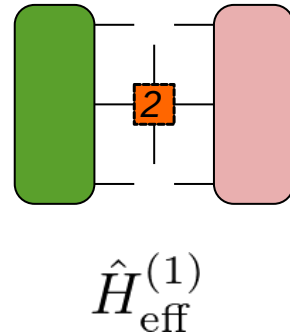
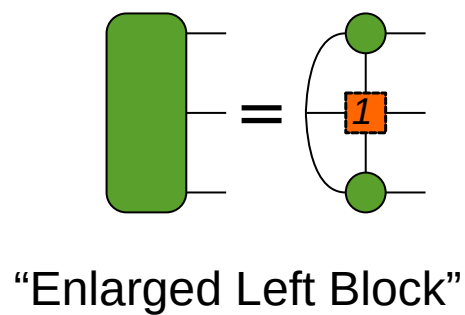


2-site forward step (1st iteration)

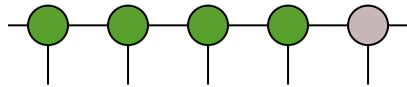


TDVP recap

1-site backward step (do it, except for right-most site)

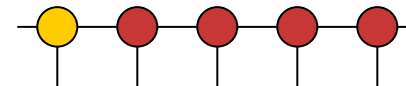


Left-to-right sweep (2-site forward, 1-site backward, except for last site)



Total time evolution up to here: $dt/2$
(L-1) forward steps, (L-2) backward steps

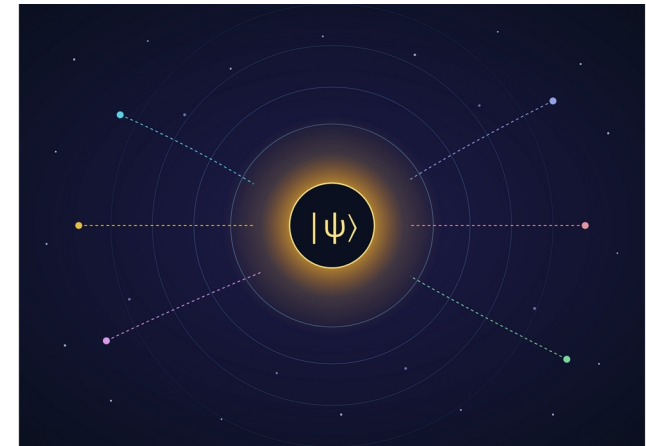
Mirror the sweep to get the wavefunction at dt :



Motivation (Open systems)

Quantum systems are never fully isolated

Engineered dissipation may open new ways to manipulate quantum systems

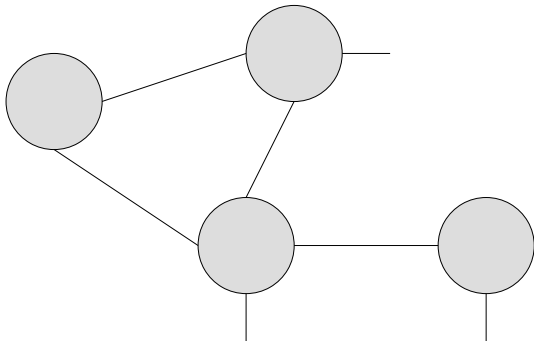


Numerical description is hard

$$\begin{aligned} |\Psi\rangle &\in \mathcal{H} \\ \rho &\in \mathcal{S}(\mathcal{H}) \end{aligned}$$

$$\dim \mathcal{S}(\mathcal{H}) = (\dim \mathcal{H})^2 - 1$$

Tensor Network methods are powerful tools for pure states



Generalize them to open systems!

How efficient is this?

Infinitely hot state is very simple:

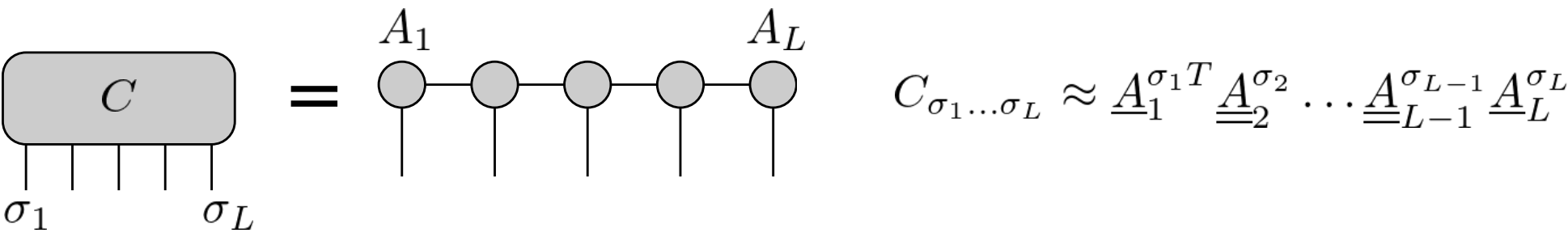
$$\rho = \frac{1}{\dim \mathcal{H}} \mathbb{I}$$

Matrix Product States

$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_1$

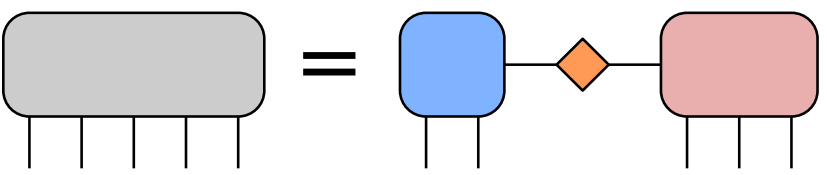
Product of small single-site Hilbert spaces

$|\Psi\rangle = \sum_{\sigma_1 \dots \sigma_L} C_{\sigma_1 \dots \sigma_L} |\sigma_1\rangle \otimes \cdots \otimes |\sigma_L\rangle$



Truncation based on Schmidt decomposition

$C_{\sigma_1 \dots \sigma_L} = \sum_{\alpha} U_{\{\sigma_1 \dots \sigma_l\} \alpha} \lambda_{\alpha} V_{\alpha \{\sigma_{l+1} \dots \sigma_L\}}$



$1 \geq \lambda_{\alpha} \geq 0$

$\sum |\lambda_{\alpha}|^2 = \langle \Psi | \Psi \rangle = 1$



L-1 SVD and truncation steps

➡

MPS

Real time simulations: the TEBD algorithm

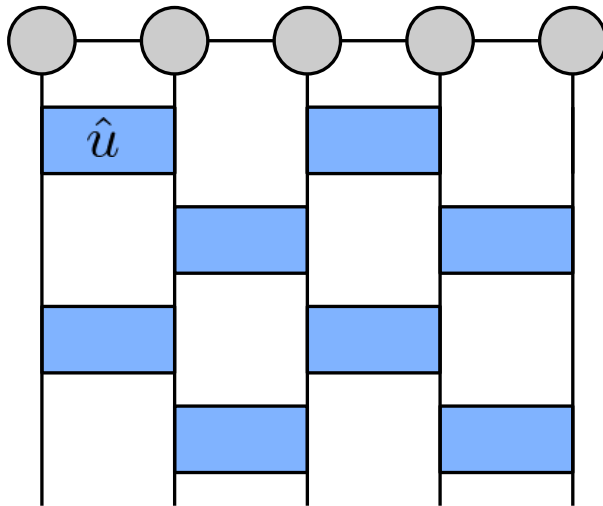
$$i\partial_t|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$$

$$|\Psi(t)\rangle = e^{-iHt}|\Psi(0)\rangle$$

1D lattice with only nearest neighbor couplings

$$\hat{H} = \sum_i \hat{h}_{i,i+1}$$

$$\hat{H} = \underbrace{\sum_k \hat{h}_{2k,2k+1}}_{\hat{H}_E} + \underbrace{\sum_k \hat{h}_{2k+1,2k+2}}_{\hat{H}_O}$$



$$\hat{u} = e^{-i\hat{h}\Delta t}$$

Example: Heisenberg spin chain

$$\hat{H} = \sum_i \vec{S}_i \cdot \vec{S}_{i+1}$$

Trotter-Suzuki approximation (1st order)

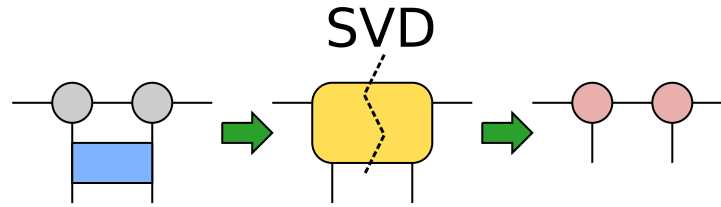
$$e^{-i\hat{H}\Delta t} = e^{-i\hat{H}_E\Delta t} e^{-i\hat{H}_O\Delta t} + O(\Delta t^2)$$

$$e^{-i\hat{H}_O\Delta t} = \prod_k e^{-i\hat{h}_{2k+1,2k+2}\Delta t}$$

$$e^{-i\hat{H}_E\Delta t} = \prod_k e^{-i\hat{h}_{2k,2k+1}\Delta t}$$

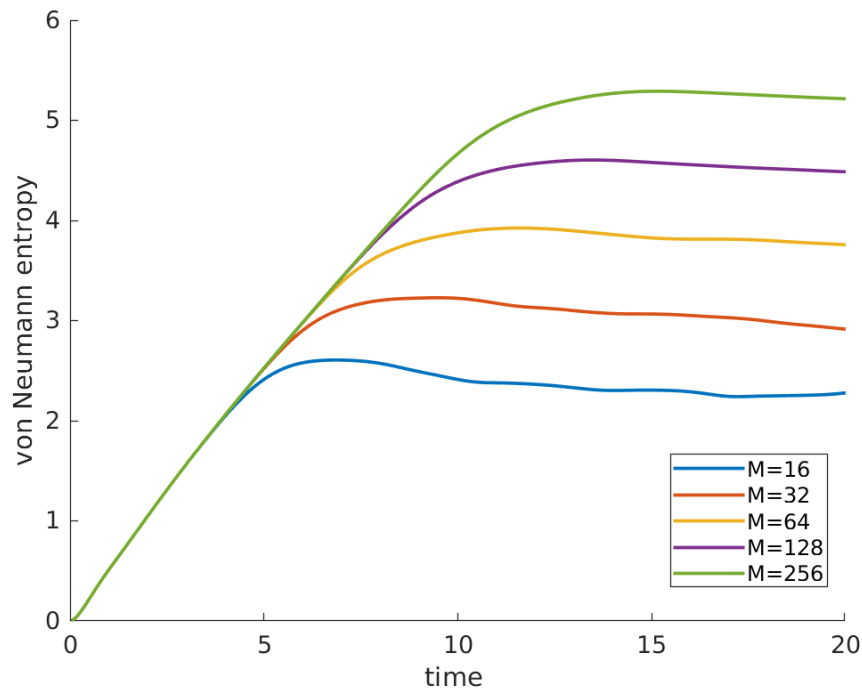
Real time simulations: the TEBD algorithm

Elementary update step:



- **Local:** only 2 tensors are modified
- Updates in the same layer can be performed parallel.

Rapidly increasing entanglement



$$S_{\text{vN}} = - \sum_{\alpha} |\lambda_{\alpha}|^2 \log |\lambda_{\alpha}|^2$$

TEBD breaks down for long times

Linear entropy growth



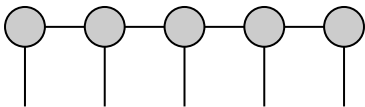
exponentially increasing
numerical cost

Doubling the bond dimension extends the
simulation frame only by some fixed (small) time

Symmetries in Matrix Product States

Abelian symmetries:

$$C_{\sigma_1 \dots \sigma_L} \approx \underline{A}_1^{\sigma_1 T} \underline{A}_2^{\sigma_2} \dots \underline{A}_{L-1}^{\sigma_{L-1}} \underline{A}_L^{\sigma_L}$$



$$\sigma_i \in \pm 0.5 \quad S_{\text{tot}}^z \Leftrightarrow \sum_i \sigma_i$$

$$\hat{H} = \sum_i \vec{S}_i \cdot \vec{S}_{i+1} \qquad [H, S_{\text{tot}}^z] = 0$$

Block structure of matrices:

- Faster tensoralgebra
- SVD sector-by-sector
- No extra algorithmic changes

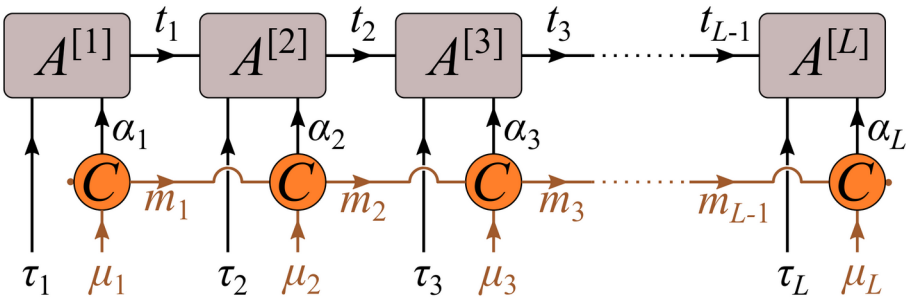
$$A_{\alpha_{i-1} \alpha_i}^{\sigma_i} \qquad \alpha_i \rightarrow \{\alpha_i, S^z(\alpha_i)\}$$

Nonzero elements only for

$$A_{\alpha_{i-1} \alpha_i}^{\sigma_i} \neq 0 \Rightarrow S^z(\alpha_i) = S^z(\alpha_{i-1}) + \sigma_i$$

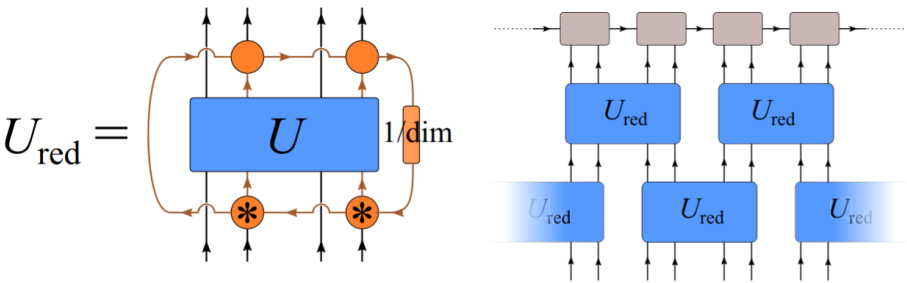
Non-Abelian symmetries:

Degenerate Schmidt weights
Two layers: Structural tensors and Clebsch-Gordans

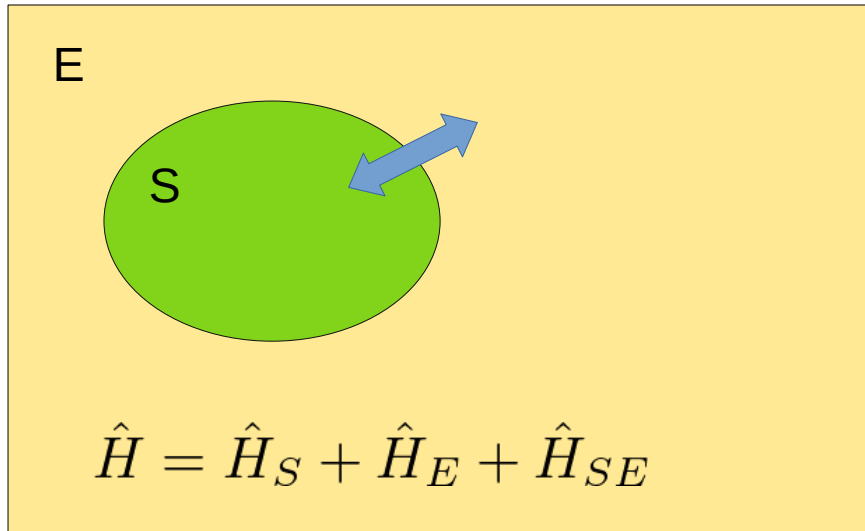


Cut the CG layer!

Modified algorithms



Open Quantum Systems



$$|\Psi\rangle = \sum_{i,j} C_{ij} |i\rangle_S \otimes |j\rangle_E$$

$$i\partial_t |\Psi\rangle = \hat{H} |\Psi\rangle$$

Density Matrix formalism (von Neumann eq.)

$$\hat{\rho} = |\Psi\rangle\langle\Psi| \quad i\partial_t \hat{\rho} = [\hat{H}, \hat{\rho}]$$

Density matrix of 'S': $\hat{\rho}_S = \text{Tr}_E \hat{\rho} = \sum_{i,i'} \underbrace{\left(\sum_j C_{ij} C_{i'j}^* \right)}_{\rho_{ii'}^{(S)} \in \text{dim}_S \times \text{dim}_S} |i\rangle\langle i'|$

Time evolution of the system?

$$\hat{\rho}_S(t) = \text{Tr}_E \hat{\rho}(t)$$

$$i\partial_t \hat{\rho}_S(t) = \text{Tr}_E [\hat{H}, \hat{\rho}(t)]$$

Full (S+E) system!



Approximation is needed

Open Quantum Systems – The Lindblad equation

$$i\partial_t \hat{\rho}_S = \underbrace{[\hat{H}_S + \hat{H}_{LS}, \rho_S]}_{\text{System Hamiltonian gets slightly modified}} + i \sum_k \Gamma_k \underbrace{\left(\hat{F}_k \hat{\rho}_S \hat{F}_k^\dagger - \frac{1}{2} \{ \hat{F}_k^\dagger \hat{F}_k, \hat{\rho}_S \} \right)}_{\text{Dissipators ("jump operators")}}$$

Non-unitary evolution

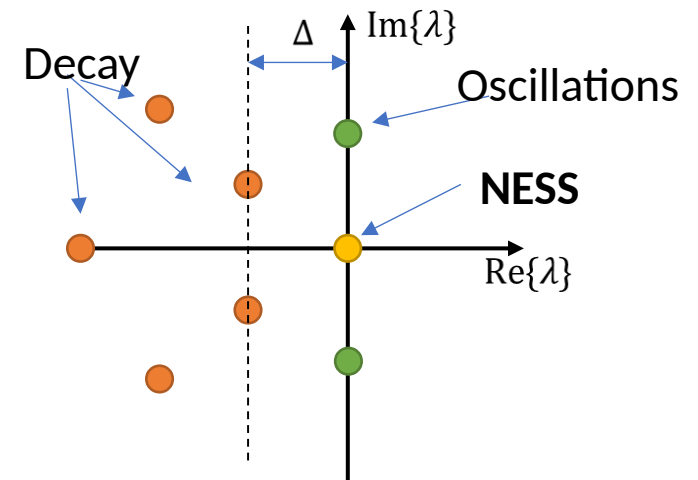
- Completely positive trace preserving (CPTP) dynamics
- Markovian dynamics (no memory)
- Temperature of the environment? Thermalization? $\longleftrightarrow$ Dissipator engineering is necessary

- $\hat{\rho}_S(t \rightarrow \infty) = \hat{\rho}_S^{\text{NESS}}$ Non-equilibrium steady state (usually unique)

- The Lindblad equation is linear

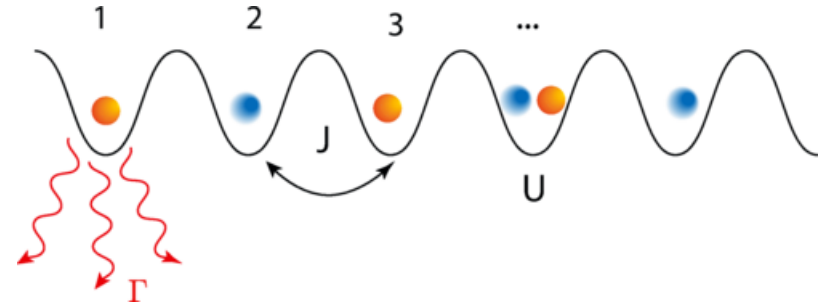


Rather straightforward generalization of TNS algorithms is possible



Hot Hubbard chain with a dissipative sink

$$H = -\frac{J}{2} \sum_{i=1}^{L-1} \sum_{\sigma=\uparrow,\downarrow} \left(c_{i\sigma}^\dagger c_{i+1\sigma} + h.c. \right) + U \sum_i n_{i\uparrow} n_{i\downarrow}$$



$$i \frac{d\rho}{dt} = [H, \rho] + i \Gamma \sum_{\sigma} \left(2 c_{1\sigma} \rho c_{1\sigma}^\dagger - \{ c_{1\sigma}^\dagger c_{1\sigma}, \rho \} \right)$$

$$\rho(t=0) = \frac{1}{4^L} \mathbb{I} \quad \text{Infinite temperature Gibbs state}$$

Motivation:

- Experimentally system
(Cold atoms in an optical lattice. Dissipation with focused laser beams)
- Highly symmetric chain: generalization of NA-MPS techniques
- Integrable chain: possibly interesting dynamics:
 - Charge transport?
 - Dependence on dissipator strength (Zeno-effect?)
 - Correlations, quantum / classical ?

C. P. Moca, M. A. Werner, O. Legeza, T. Prosen, M. Kormos, and G. Zarand,
Physical Review B 105, 195144 (2022).

Symmetries in the Lindblad-equation

$$i\partial_t \rho = L[\rho]$$

$$U(g)L[\rho]U^\dagger(g) = L[U(g)\rho U^\dagger(g)] \quad \forall g \in \mathcal{G}$$

$$i\frac{d\rho}{dt} = [H, \rho] + i\Gamma \sum_{\sigma} (2c_{1\sigma}\rho c_{1\sigma}^\dagger - \{c_{1\sigma}^\dagger c_{1\sigma}, \rho\})$$

U(1) charge “conservation” $U = e^{i\varphi N_{\text{tot}}}$

$$[N_{\text{tot}}, H] = 0$$

$$U[H, \rho]U^\dagger = [H, U\rho U^\dagger]$$

$$[N_{\text{tot}}, c_{1\sigma}^\dagger c_{1\sigma}] = 0$$

$$U\{c_{1\sigma}^\dagger c_{1\sigma}, \rho\}U^\dagger = \{c_{1\sigma}^\dagger c_{1\sigma}, U\rho U^\dagger\}$$

$$Uc_{1\sigma} = c_{1\sigma}Ue^{-i\varphi}$$

$$Uc_{1\sigma}\rho c_{1\sigma}^\dagger U^\dagger = c_{1\sigma}U\rho U^\dagger c_{1\sigma}^\dagger$$

$$c_{1\sigma}^\dagger U^\dagger = U^\dagger c_{1\sigma}^\dagger e^{+i\varphi}$$

Consequence: $[\rho(t=0), N_{\text{tot}}] = 0 \Rightarrow [\rho(t), N_{\text{tot}}] = 0 \quad \forall t$

Remark: $[H, U] = 0$

And dissipators have ‘good’ quantum numbers

$$Uc_{1\sigma}U^\dagger = e^{-i\varphi}c_{1\sigma}$$

“Charge of operator” is -1

Symmetries in the Lindblad-equation

$$i\partial_t \rho = L[\rho]$$

$$i\frac{d\rho}{dt} = [H, \rho] + i\Gamma \sum_{\sigma} (2c_{1\sigma}\rho c_{1\sigma}^{\dagger} - \{c_{1\sigma}^{\dagger}c_{1\sigma}, \rho\})$$

SU(2) spin “conservation” $U(\vec{n}, \varphi) = e^{i\vec{n}\vec{S}_{\text{tot}}}$

$$\begin{pmatrix} c_{1,\downarrow} \\ -c_{1,\uparrow} \end{pmatrix} : \text{spin } \frac{1}{2} \text{ tensor operator} \quad U(g) \begin{pmatrix} c_{1,\downarrow} \\ -c_{1,\uparrow} \end{pmatrix} U(g)^{\dagger} = \underline{\underline{R}}_{1/2}(g) \begin{pmatrix} c_{1,\downarrow} \\ -c_{1,\uparrow} \end{pmatrix}$$

Spin half representation
of the SU(2) group

$$U(g)L[\rho]U^{\dagger}(g) = L[U(g)\rho U^{\dagger}(g)] \quad \forall g \in \mathcal{G} \quad \textbf{Requirement: } \Gamma_{\uparrow} = \Gamma_{\downarrow} = \Gamma$$

$$[\rho(t=0), \vec{S}_{\text{tot}}] = 0 \Rightarrow [\rho(t), \vec{S}_{\text{tot}}] = 0 \quad \forall t$$

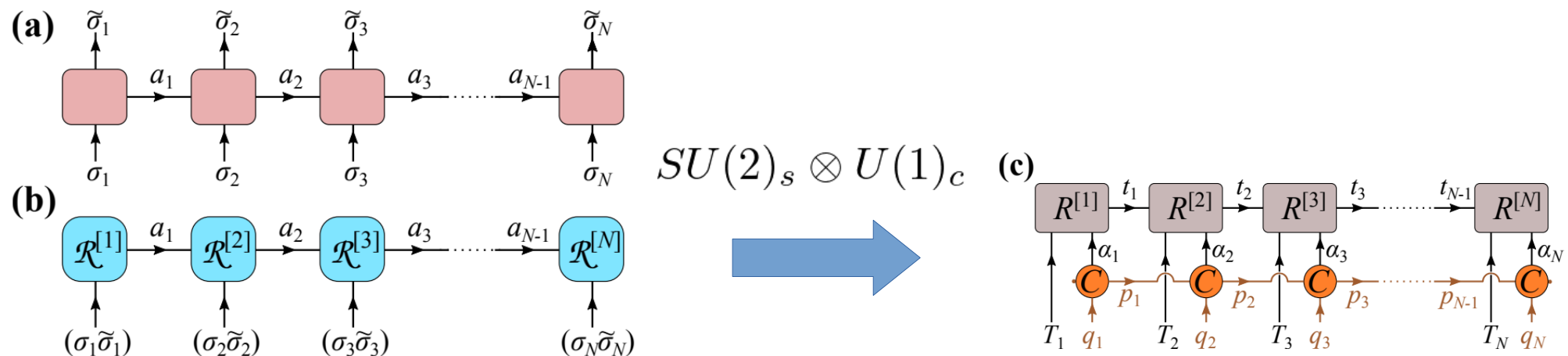
Block diagonal density matrix remains
block diagonal

Towards MPS simulation: vectorization

$$\hat{\rho} = \sum_{i, \tilde{i}} \rho_{i\tilde{i}} |i\rangle \langle \tilde{i}| \quad \longrightarrow \quad |\rho\rangle = \sum_{i\tilde{i}} \rho_{i\tilde{i}} |i\rangle \otimes |\tilde{i}\rangle$$

$$i\partial_t \hat{\rho} = [\hat{H}, \hat{\rho}] + i \sum_k \Gamma_k \left(\hat{F}_k \hat{\rho} \hat{F}_k^\dagger - \frac{1}{2} \{ \hat{F}_k^\dagger \hat{F}_k, \hat{\rho} \} \right)$$

$$\mathbb{L} = H \otimes \mathbb{I} - \mathbb{I} \otimes H^* + i \sum_k \Gamma_k \left(F_k \otimes F_k^* - \frac{1}{2} F_k^\dagger F_k \otimes \mathbb{I} - \frac{1}{2} \mathbb{I} \otimes (F_k^\dagger F_k)^* \right)$$



The vectorized Hubbard-Lindbladian

What is needed to use NA-MPS framework?

Local (vectorized) operator-space organized into operator multiplets

- Automatized method based on the Wigner-Eckart theorem?
- By hand (superfermions in a super Fock space)

A. A. Dzhioev and D. S. Kosov, J. Chem. Phys. **134**, 044121 (2011)

$$c_{i\sigma}^\dagger, c_{i,\sigma} \text{ \& \> } \tilde{c}_{i\sigma}^\dagger, \tilde{c}_{i,\sigma}$$

‘ket’ fermions

‘bra’ fermions

$$\begin{aligned} \mathbb{L} = & -\frac{J}{2} \sum_{\sigma} \sum_{i=1}^{L-1} (c_{i\sigma}^\dagger c_{i+1\sigma} + h.c.) + U \sum_i n_{i\uparrow} n_{i\downarrow} \\ & + \frac{J}{2} \sum_{\sigma} \sum_{i=1}^{L-1} (\tilde{c}_{i\sigma}^\dagger \tilde{c}_{i+1\sigma} + h.c.) - U \sum_i \tilde{n}_{i\uparrow} \tilde{n}_{i\downarrow} \\ & + 2i\Gamma \sum_{\sigma} c_{1\sigma} \tilde{c}_{1\sigma} - i\Gamma \sum_{\sigma} (\tilde{c}_{1\sigma}^\dagger \tilde{c}_{1\sigma} - c_{1\sigma}^\dagger c_{1\sigma}) . \end{aligned}$$

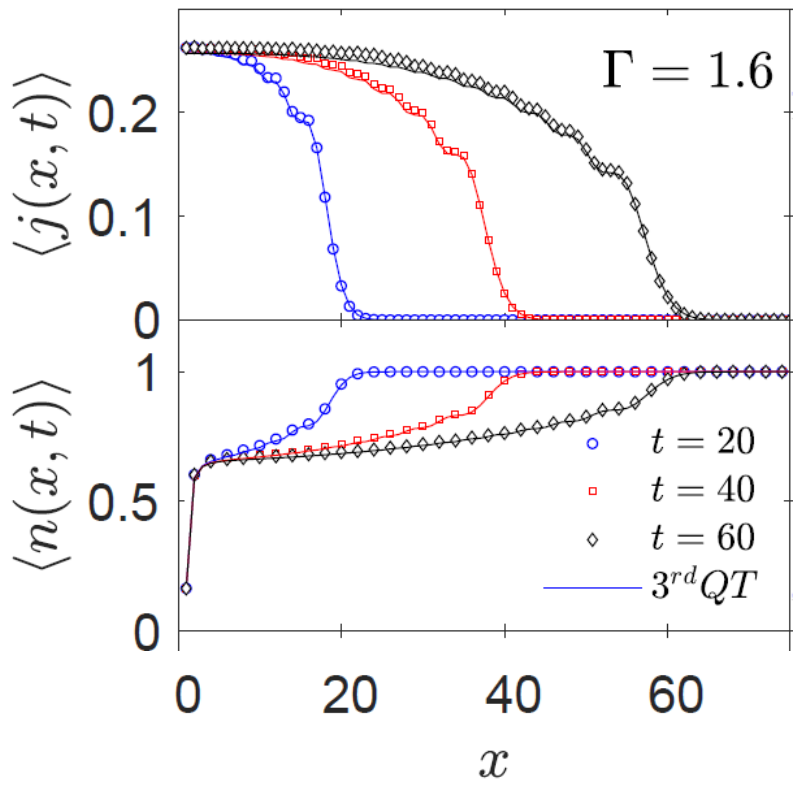
Conservation laws:

$$\mathbb{N}_j = N_j \otimes I - I \otimes N_j \qquad \text{Super charge}$$

$$\mathbb{S}_j^\alpha = S_j^\alpha \otimes I - I \otimes S_j^{\alpha*} \qquad \text{Super spin}$$

$\Gamma = (\mathbb{S}, \mathbb{N})$	$\dim(\Gamma)$	T	states
$(0, -2)$	1	1	$\tilde{c}_\uparrow^\dagger \tilde{c}_\downarrow^\dagger 0\rangle$
$(0, 0)$	1	1	$ 0\rangle$
		2	$\frac{1}{\sqrt{2}}(c_\uparrow^\dagger \tilde{c}_\uparrow^\dagger - c_\downarrow^\dagger \tilde{c}_\downarrow^\dagger) 0\rangle$
		3	$c_\uparrow^\dagger c_\downarrow^\dagger \tilde{c}_\uparrow^\dagger \tilde{c}_\downarrow^\dagger 0\rangle$
$(0, 2)$	1	1	$c_\uparrow^\dagger c_\downarrow^\dagger 0\rangle$
$(\frac{1}{2}, -1)$	2	1	$\{\tilde{c}_\uparrow^\dagger 0\rangle, \tilde{c}_\downarrow^\dagger 0\rangle\}$
		2	$\{c_\uparrow^\dagger \tilde{c}_\uparrow^\dagger \tilde{c}_\downarrow^\dagger 0\rangle, c_\downarrow^\dagger \tilde{c}_\uparrow^\dagger \tilde{c}_\downarrow^\dagger 0\rangle\}$
$(\frac{1}{2} ; 1)$	2	1	$\{c_\uparrow^\dagger 0\rangle, c_\downarrow^\dagger 0\rangle\}$
		2	$\{c_\uparrow^\dagger c_\downarrow^\dagger \tilde{c}_\downarrow^\dagger 0\rangle, c_\uparrow^\dagger c_\downarrow^\dagger \tilde{c}_\uparrow^\dagger 0\rangle\}$
$(1, 0)$	3	1	$\{ c_\downarrow^\dagger \tilde{c}_\uparrow^\dagger 0\rangle$
			$\frac{1}{\sqrt{2}}(c_\uparrow^\dagger \tilde{c}_\uparrow^\dagger + c_\downarrow^\dagger \tilde{c}_\downarrow^\dagger) 0\rangle$ $c_\uparrow^\dagger \tilde{c}_\downarrow^\dagger 0\rangle \}$

Ballistic charge/current front, $U=0$



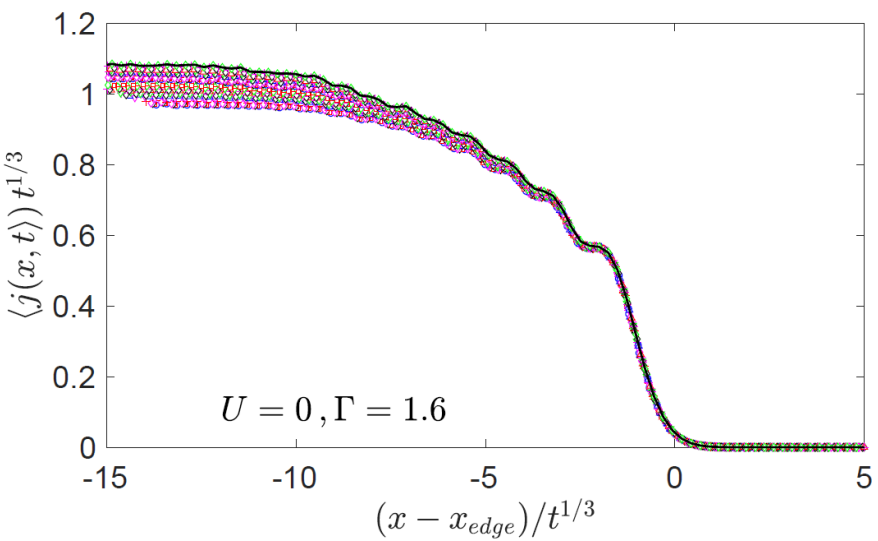
- Ballistic front propagation
 $v_{\text{edge}} = J \longleftrightarrow$ Max. velocity of fermions
- Universal current profile at the edge
$$j(x, t) \sim \frac{1}{t^{1/3}} \Phi\left(\frac{x - x_{\text{edge}}}{t^{1/3}}\right)$$

Non-interacting Hamiltonian & simple dissipator

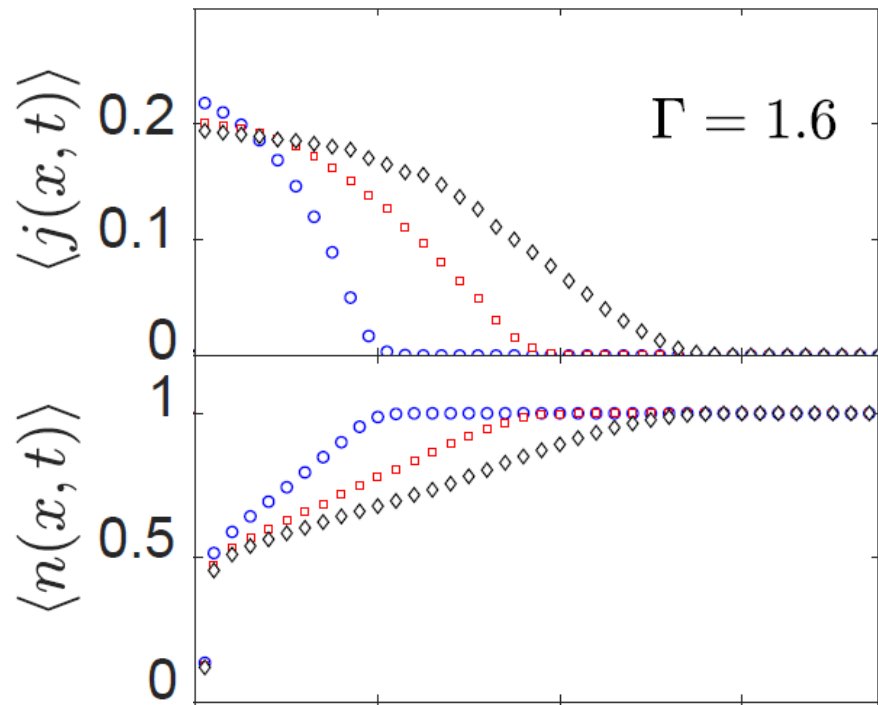


Quadratic Lindbladian

- Exactly solvable
- 3^{rd} quantization method
T. Prosen, New Journal of Physics 10, 043026 (2008)



Ballistic charge/current front, $U=1$

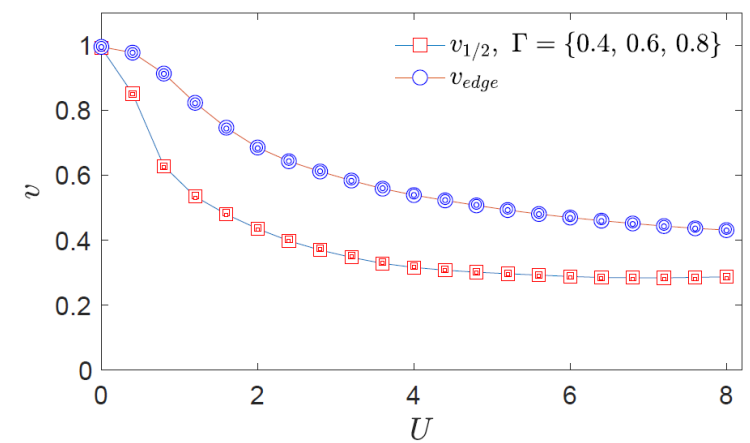
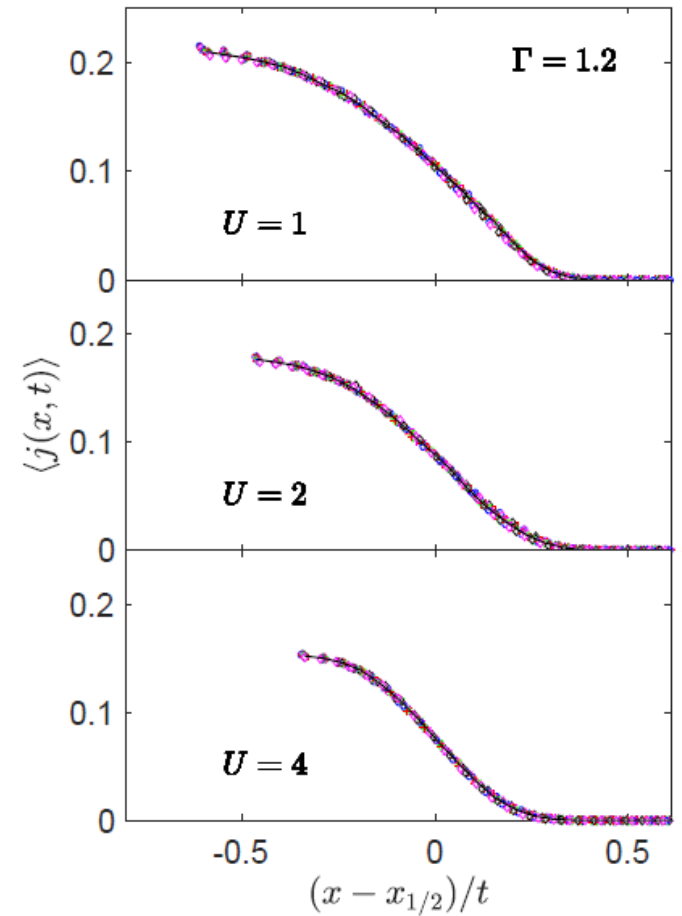


- Ballistic front propagation, but

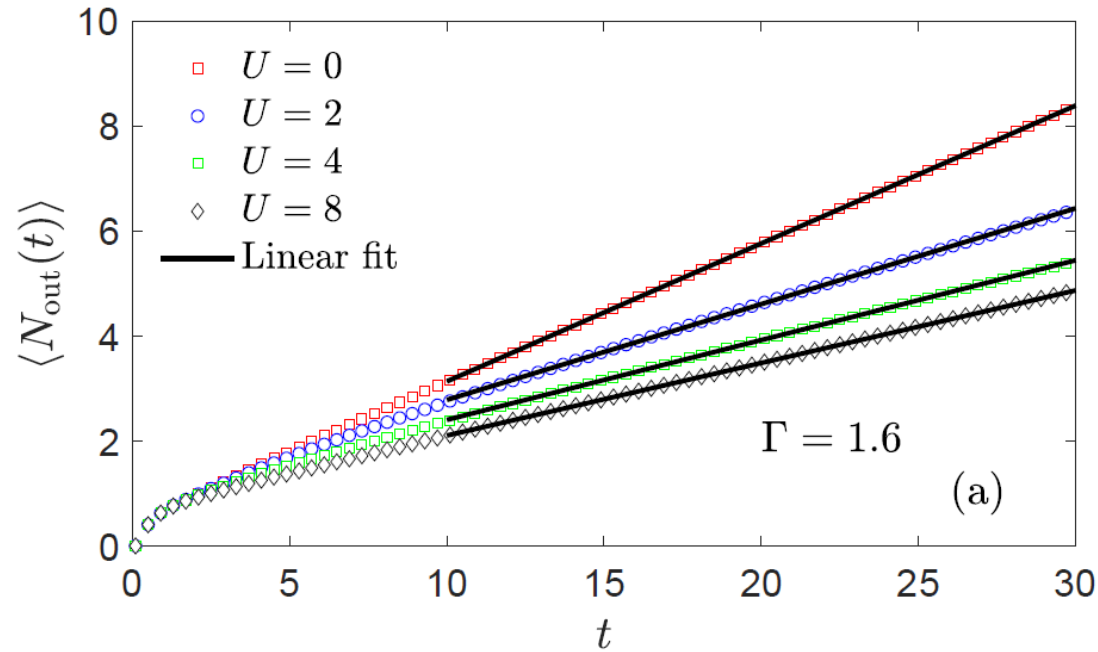
$$v_{\text{edge}} < J \quad \longleftrightarrow \quad \text{Interactions slow down front propagation}$$

- Universal current profile at the edge

$$j(x, t) \sim \Xi\left(U, \frac{x - x_{\text{edge}}}{t}\right)$$

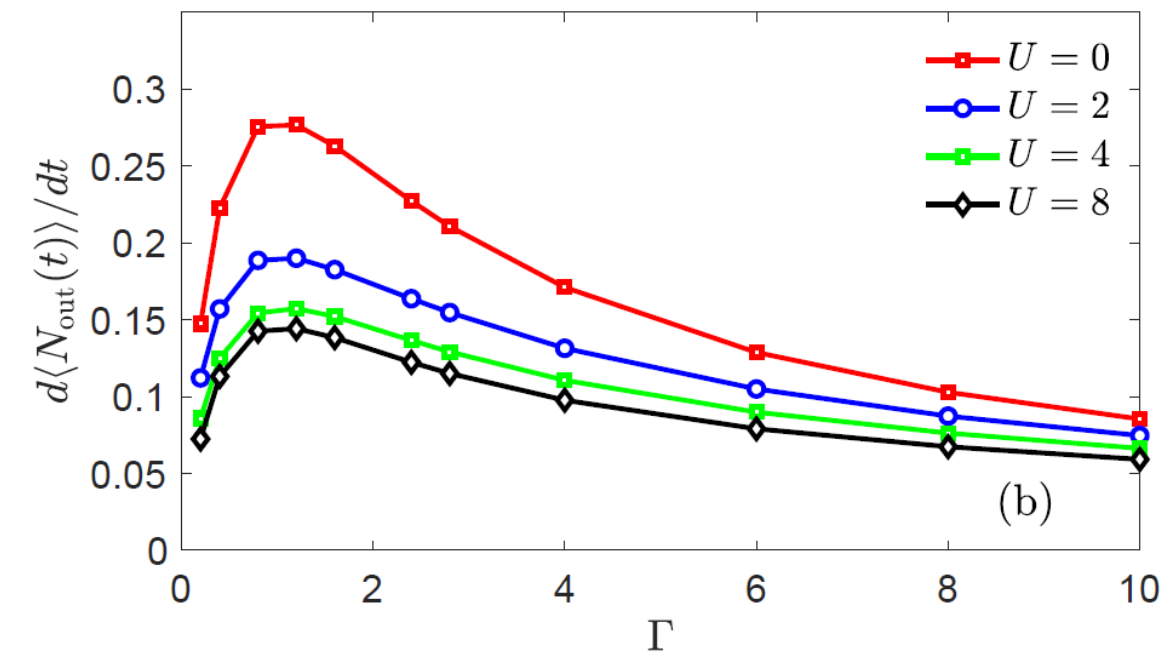


Role of dissipator strength – Zeno effect



$$j(x \ll v_F t, t) \sim \text{const}$$

$$j = -\frac{dN_{out}}{dt}$$

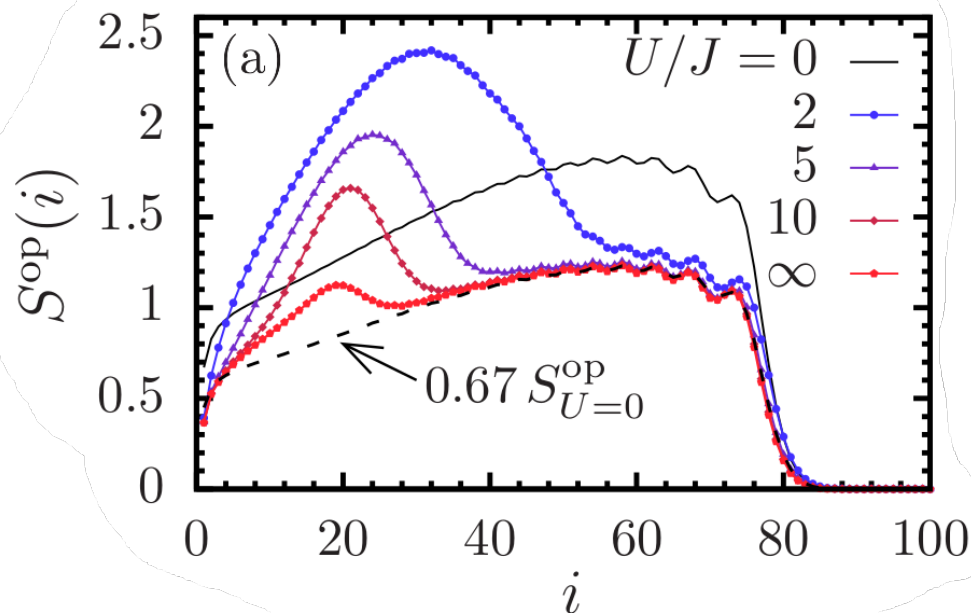
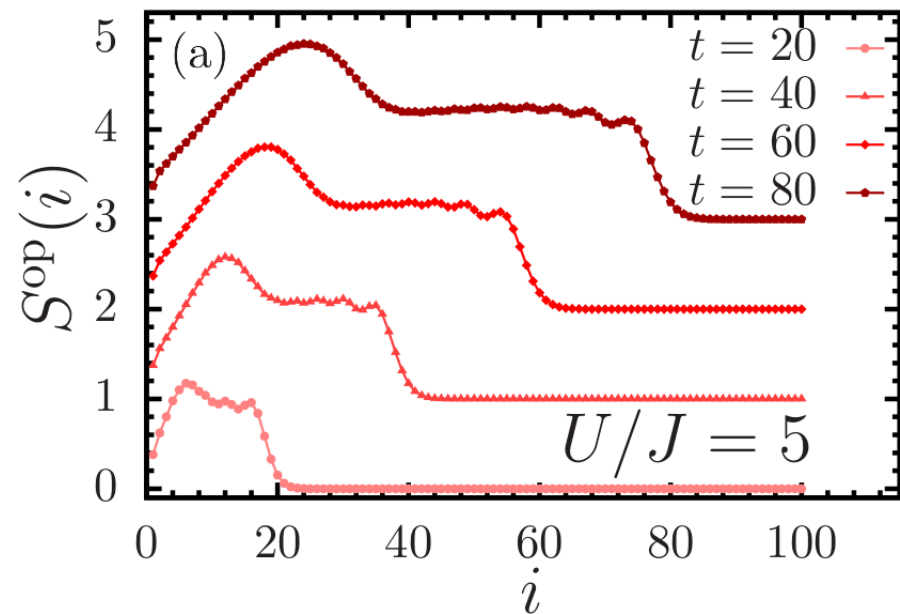


Stronger coupling to the environment leads to weaker current



Quantum Zeno effect

“Operator entanglement” – double front



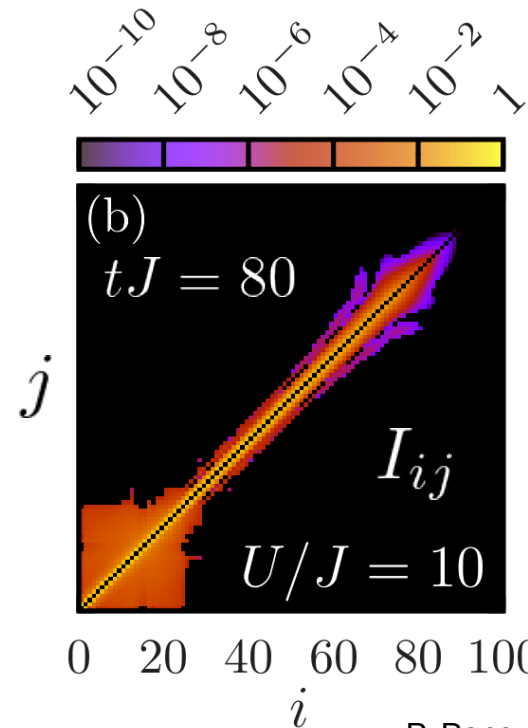
Schmidt decomposition

$$A : [1, 2, \dots, i] \quad B : [i + 1, i + 2, \dots]$$

$$|\rho\rangle = \mathcal{N} \sum_{\alpha} \lambda_{\alpha} |\alpha\rangle_A \otimes |\alpha\rangle_B$$

“Operator entropy”

$$S^{\text{op}} = - \sum_{\alpha} \lambda_{\alpha}^2 \ln(\lambda_{\alpha}^2)$$

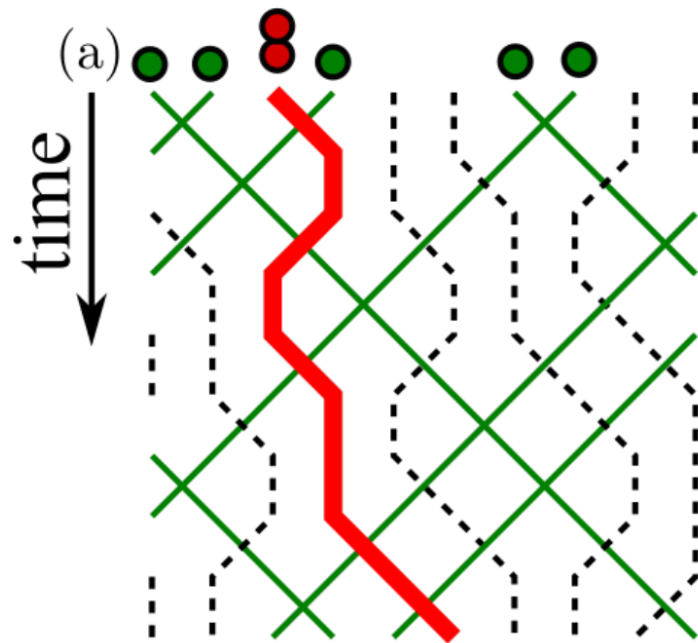


Mutual information

$$I_{ij} = S_i + S_j - S_{ij}$$

1- and 2-site physical entropy

Reversible Cellular Automaton model



$$U \rightarrow \infty$$

- **Energy conservation:** creation and annihilation of doublons is forbidden
 - No direct hopping of doublons
 - Doublon 'motion' at doublon-singlon collisions

Discrete dynamics instead of continuous evolution



Reversible Cellular Automaton model (RCA)

T. Gombor and B. Pozsgay, SciPost Phys. 12, 102 (2022).

Initial state ($T=\infty$): fully random (uniform distribution)

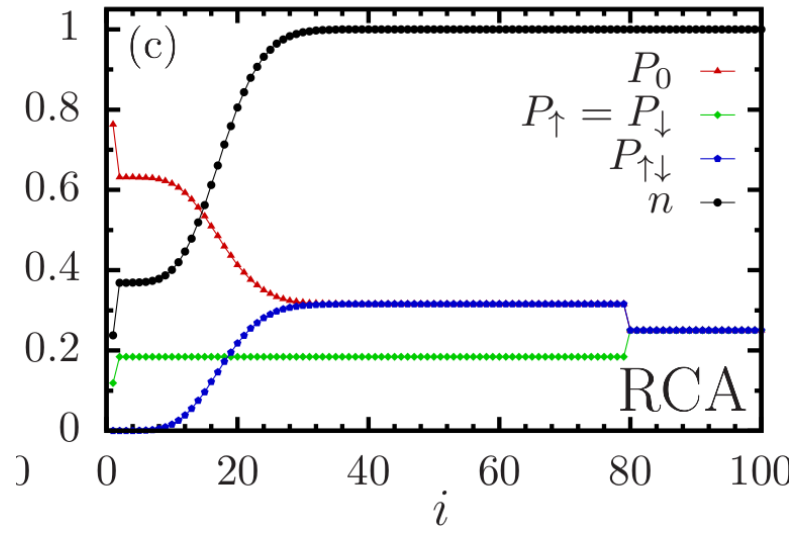
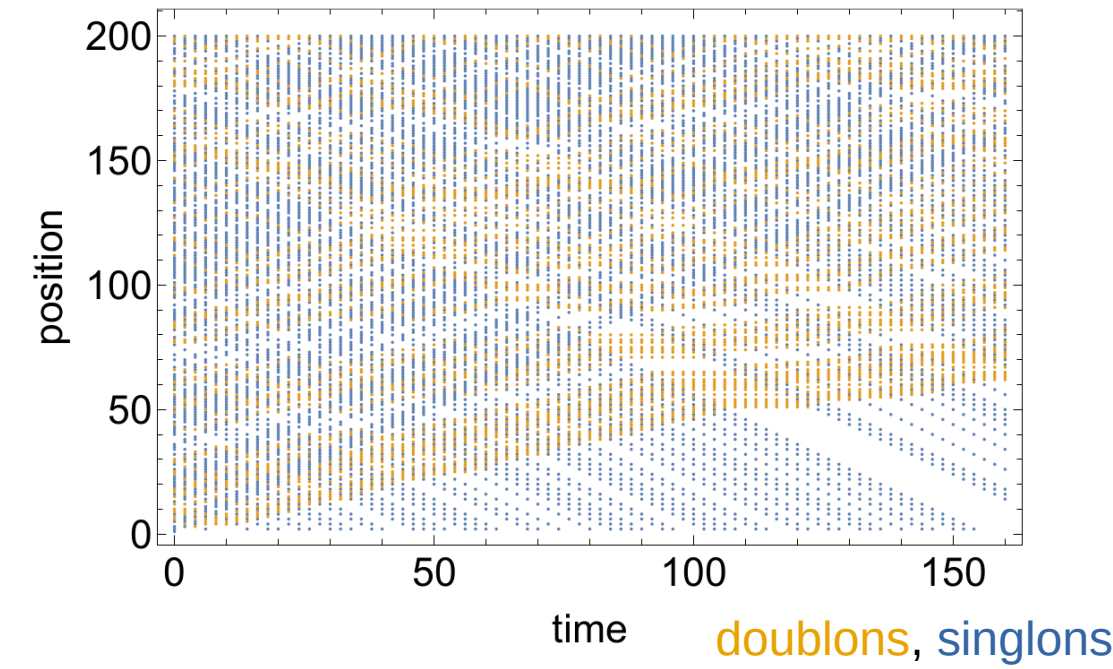
Particle sink: Particles are removed from the first site (removal probability: $0 \leq q \leq 1$)



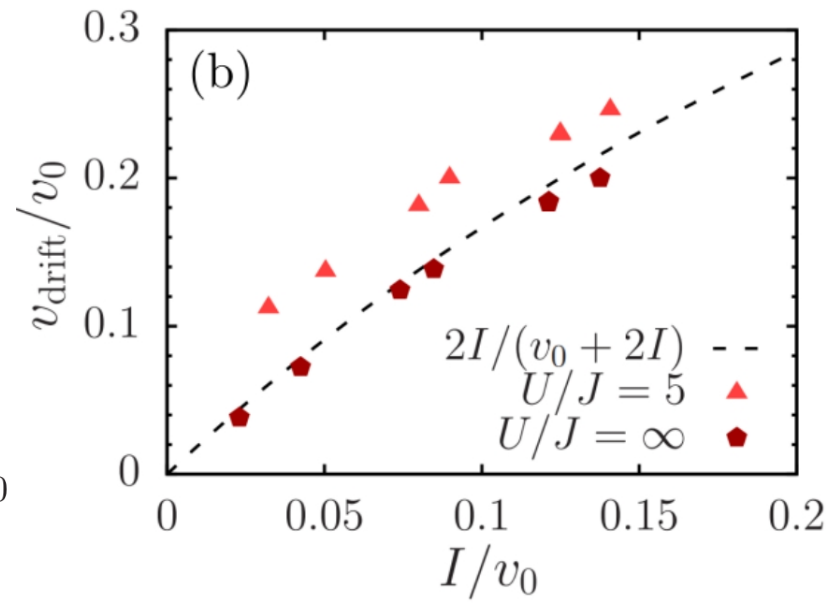
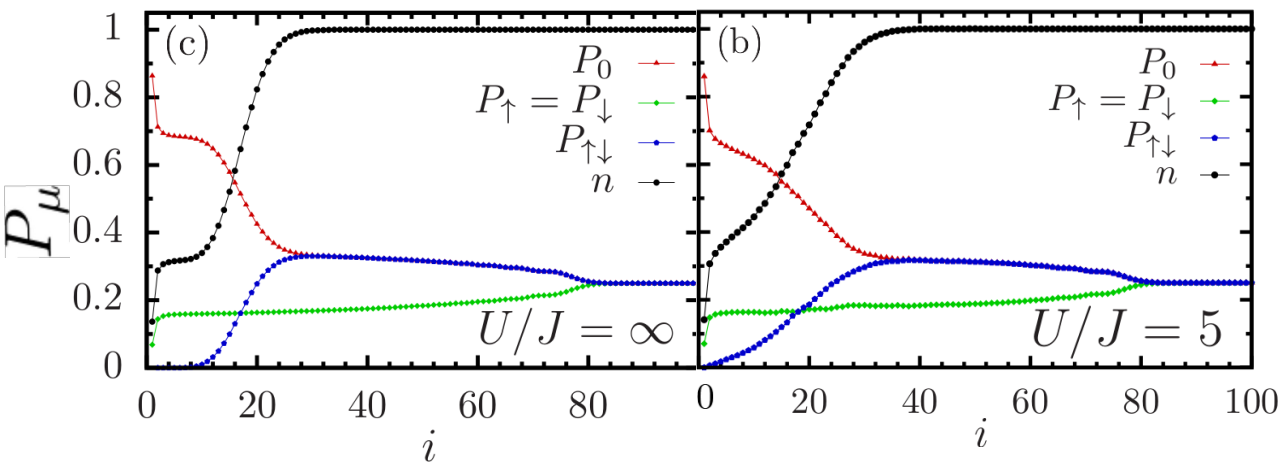
Solution strategies:

- Monte Carlo sampling of initial states, direct simulation of dynamics
- Modified MPS simulation: discrete 2-site gates, diagonal density matrix

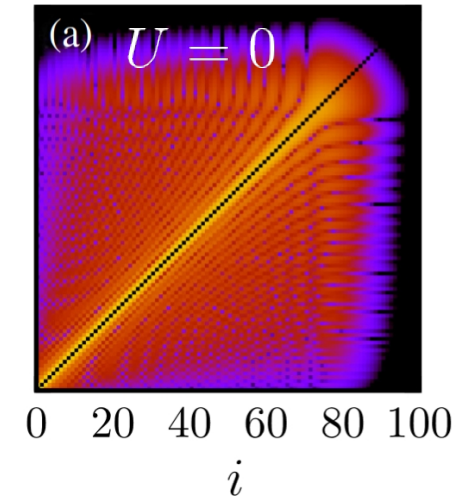
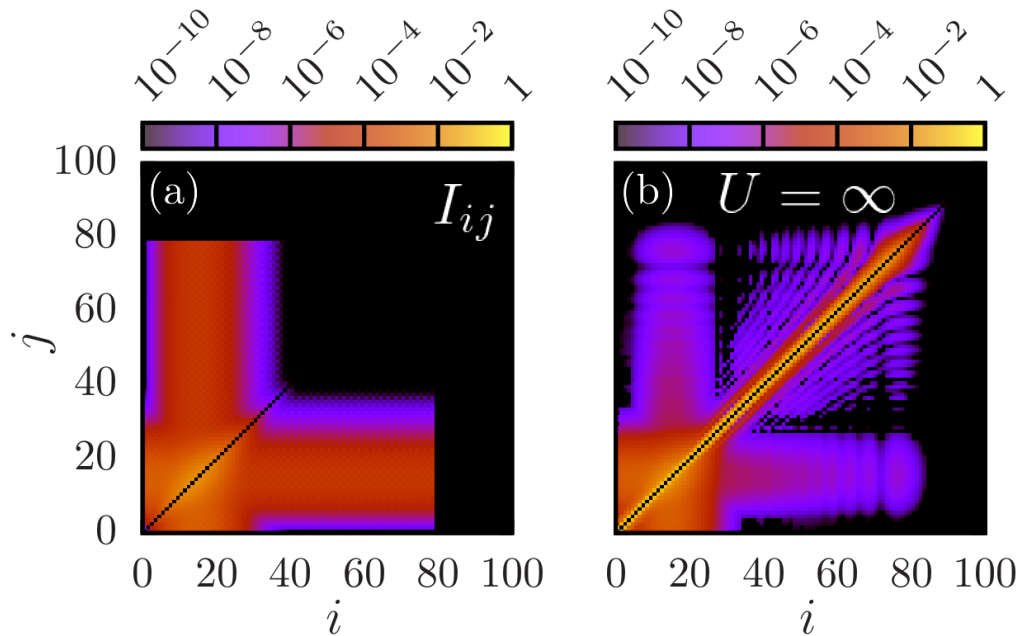
Double front in the RCA model



Hubbard model simulation results:



Fast front: “quantum information jet”



Kumar’s mapping (B. Kumar, Phys. Rev. B 79, 155121 (2009))

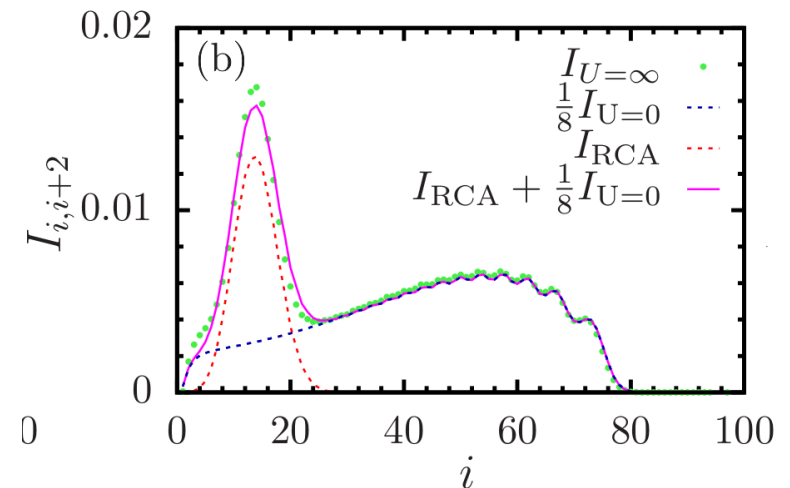
$$\{|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle\} \Rightarrow \{|0\uparrow\rangle, |1\uparrow\rangle, |1\downarrow\rangle, |0\downarrow\rangle\}$$

Spinless fermion over a spin background

$$H_{U=\infty} = -\frac{J}{2} \sum_i \left(f_i^\dagger f_{i+1} + h.c. \right) X_{i,i+1}^s$$

- Exactly solvable dynamics

- 2-site mutual information is exponentially suppressed $I_{ij}^{U=\infty} \simeq \frac{I_{ij}^{U=0}/2}{4^{|i-j|-1}}$



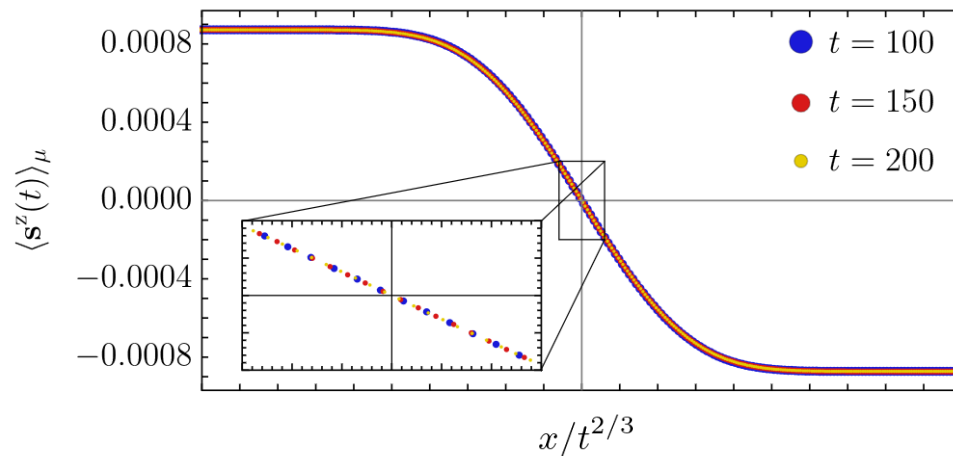
Kardar Parisi Zhang scaling

First observed in SU(2) symmetric spin chains (Heisenberg chain)

M Ljubotina, M Žnidarič, T Prosen, Phys. Rev. Lett. 122, 210602 (2019)

$$\rho(t=0) \propto \rho_\mu = (e^{\mu s^z})^{\otimes L/2} \otimes (e^{-\mu s^z})^{\otimes L/2}$$

Initial state with small imbalance



- Universal super-diffusive melting of the domain wall.
- Dynamical exponent resembles classical KPZ equation

Generalization to Hubbard chain:

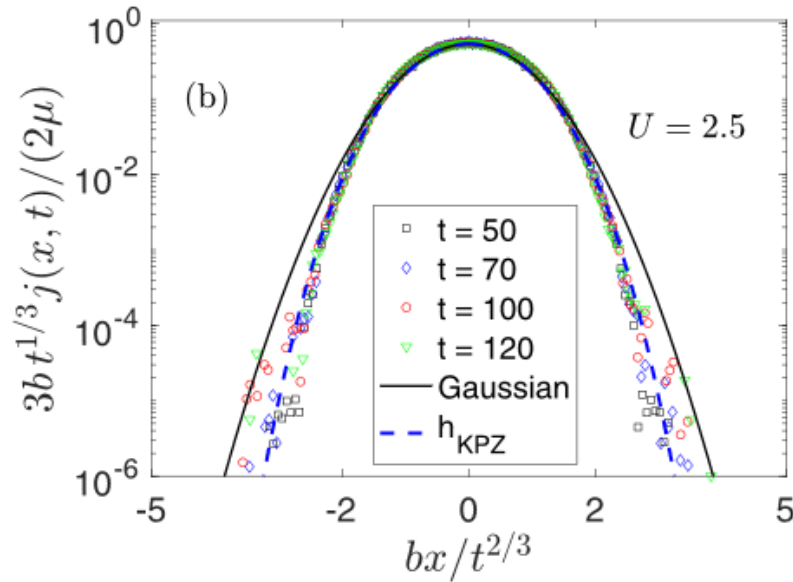
$$H = -\frac{J}{2} \sum_{i=1}^{L-1} \sum_{\sigma=\uparrow,\downarrow} \left(c_{i\sigma}^\dagger c_{i+1\sigma} + h.c. \right) + U \sum_i n_{i\uparrow} n_{i\downarrow} + \sum_{i,\sigma} \mu_\sigma c_{i\sigma}^\dagger c_{i\sigma}$$

$$SU(2)_s \times U(1)_c$$

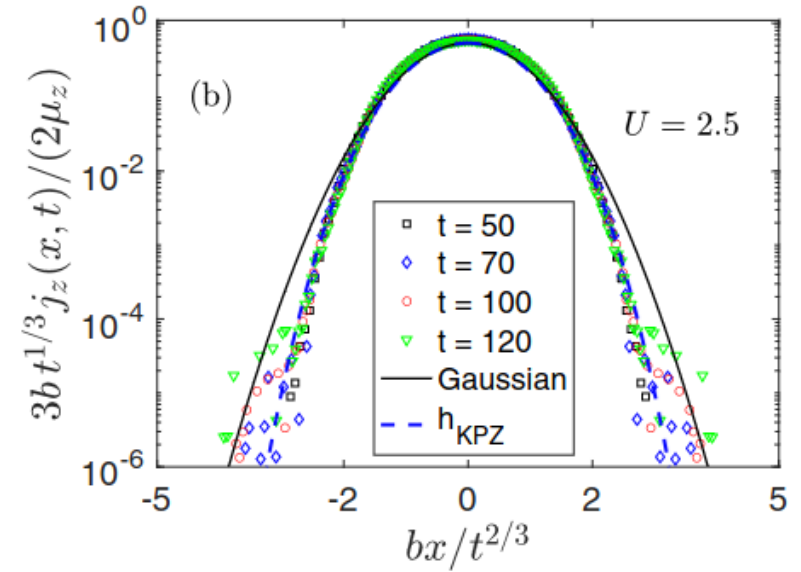
At half filling: $SU(2)_s \times SU(2)_c$

KPZ scaling: the role of symmetries

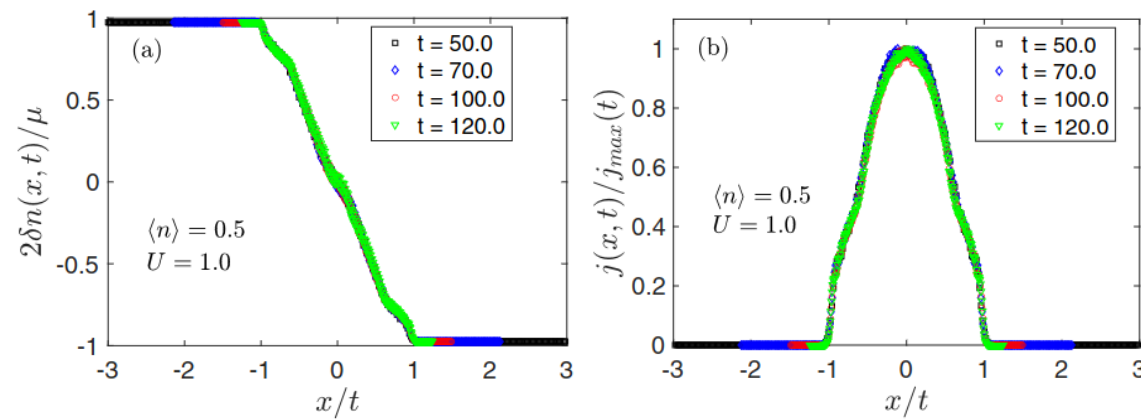
Half filling, charge transport



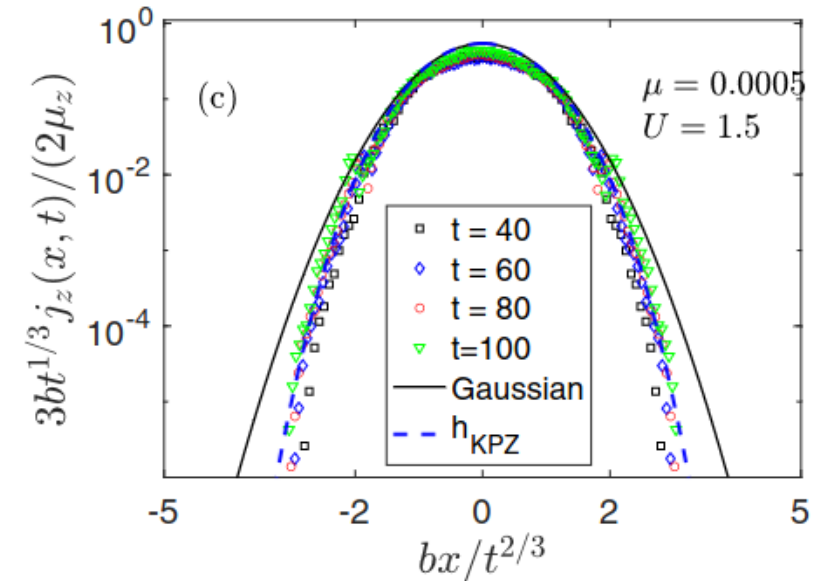
Half filling, spin transport



Quarter filling, charge transport



Quarter filling, spin transport



Summary

- Efficient NA-MPS framework for the Lindblad equation.
- Ballistic charge transport in the Hubbard chain.
Breakdown of classical transport theory
- Quantum Zeno effect (stronger coupling – lower current)
- Front separation, classical RCA model and quantum information jet
- Super-diffusive KPZ scaling at $SU(2)$ symmetric points.

Open questions:

- Fast front is understood for $U \rightarrow \infty$, but is also present at finite U .
- Higher dimensional or non-integrable models?
- More efficient MPS simulation using fermionic mode transformation?